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### Transcritical CO.SUB.2.pulverization

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#### Abstract

Methods and apparatus for comminution of solid materials. Solid materials are introduced into a first pressure vessel. A working fluid is provided in the first pressure vessel at an operating pressure and an operating temperature. The working fluid is allowed to permeate into the solid materials. The working fluid may permeate into the solid materials while in a supercritical fluid phase. The working fluid is rapidly expanded to thereby create fractured solid materials from the solid materials. Rapidly expanding the working fluid may comprise causing a transition of the working fluid from a supercritical phase with a first density to a subcritical vapour phase with a second density.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of Patent Cooperation Treaty (PCT) application No. PCT/CA2021/051407 having an international filing date of 6 Oct. 2021, which in turn claims priority from, and for the purposes of the United States the benefit of 35 USC § 119 in respect of, U.S. application No. 63/088,273 filed 6 Oct. 2020. All of the applications in this paragraph are hereby incorporated herein by reference.

### TECHNICAL FIELD

(1) This invention relates generally to methods and systems for comminution of solid materials and in particular to methods and systems for comminution of ore-bearing or aggregate rock.

### BACKGROUND

(2) In mineral processing, comminution is employed to reduce the size of ore-bearing rocks. Comminution may facilitate further processing of ore-bearing rocks by reducing their size, increasing their surface area and freeing useful materials from undesirable materials in which they may be embedded.

- (3) Comminution typically includes one or more of crushing, blasting, cutting, vibrating and grinding. In some cases, rocks are crushed to achieve a more manageable size for grinding. Crushing may be accomplished by compressing rocks against rigid surfaces, or by impacting rock against surfaces in a constrained motion path. Crushing is often performed in several stages. There are a number of crushers available such as jaw, gyratory, cone, roll, and impact crushers.
- (4) Crushing may handle rocks as large as 150 cm in diameter and may reduce such rocks down to approximately 2 cm or 5 mm fragments which are then reduced to fine particles through grinding.
- (5) Traditional forms of crushing and grinding may require large amounts of energy. For example, in some cases, to achieve particle sizes on the scale of 100  $\mu\text{m}$ , crushing and grinding requires between 10 kWh and 30 kWh per tonne of rock. Some estimates suggest that comminution in the field of mineral processing consumes up to 4% of global electrical energy and about 50% of the energy of each mine site.
- (6) There is a general desire to reduce the energy required for comminution of solid materials such as, but not limited to, ore-bearing rock without reducing the effectiveness of the comminution process.
- (7) Comminution of solid materials often causes undesirable wear on the machines involved. For example, the steel balls and liners of grinders may need to be replaced regularly due to undesirable wear.
- (8) There is a general desire to reduce the maintenance required of machines employed for comminution of solid materials, such as, but not limited to, ore-bearing rock without reducing the effectiveness of the comminution process.
- (9) The foregoing examples of the related art and limitations related thereto are intended to be illustrative and not exclusive. Other limitations of the related art will become apparent to those of skill in the art upon a reading of the specification and a study of the drawings.

#### SUMMARY

- (10) One aspect of the invention provides a method for comminution of solid materials. The method may comprise introducing solid materials into a first pressure vessel, providing a working fluid in the first pressure vessel at an operating pressure and an operating temperature, allowing the working fluid to permeate into the solid materials, and rapidly expanding the working fluid to thereby create fractured solid materials from the solid materials.
- (11) In some embodiments, the working fluid is allowed to permeate into the solid materials while in a supercritical fluid phase. In some embodiments, the working fluid is allowed to permeate into the solid materials while in a subcritical fluid (e.g. liquid, vapour or gas) phase.
- (12) In some embodiments, the operating temperature is greater than a critical temperature of the working fluid. In some embodiments, the operating pressure is greater than a critical pressure of the working fluid. In some embodiments, the operating temperature is lower than a critical temperature of the working fluid. In some embodiments, the operating pressure is lower than a critical pressure of the working fluid.
- (13) In some embodiments, rapidly expanding the working fluid comprises reducing the pressure of the working fluid from the operating pressure to a second pressure lower than the operating pressure. In some embodiments, the second pressure is below a critical pressure of the working fluid.
- (14) In some embodiments, the method comprises reducing the pressure of the working fluid from the operating pressure to the second pressure over a time period of 30 ms or less (e.g. in the range of about 1 ms to about 30 ms).
- (15) In some embodiments, rapidly expanding the working fluid comprises causing a transition of the working fluid from a supercritical phase with a first density to a subcritical vapour phase with a second density. In some embodiments, rapidly expanding the working fluid comprises causing a transition of the working fluid from a liquid or dense fluid phase with a first density to a gas or vapour phase with a second density. In some embodiments, the first density is at least 2 times

greater than the second density. In some embodiments, the first density is at least 10 times greater than the second density. In some embodiments, the first density is at least 100 times greater than the second density.

(16) In some embodiments, allowing the working fluid to permeate into the solid materials comprises maintaining the working fluid at the operating temperature and the operating pressure within the first pressure vessel for a period of time. The period of time may, for example, be in the range of approximately 1 millisecond to approximately 60 seconds. In some applications it can be beneficial to maintain the working fluid at the operating temperature and the operating pressure within the first pressure vessel for a longer period such as a period in the range of approximately 1 minute to approximately 60 minutes.

(17) In some embodiments, the solid materials comprise porous ore-bearing rock. In some embodiments, the working fluid comprises CO<sub>2</sub>.

(18) In some embodiments, the operating temperature is greater than 0.0° C. In some embodiments, the operating temperature is between about 0.0° C. and 31.0° C. In some embodiments, the operating temperature is greater than 31.0° C. In some embodiments, the operating temperature is between 31.1° C. and 100.0° C. In some embodiments, the operating pressure is greater than 5.2 bar. In some embodiments, the operating pressure is between about 5.2 bar and 73.8 bar. In some embodiments, the operating pressure is greater than 73.8 bar. In some embodiments, the operating pressure is in the range of 73.9 bar to 200 bar or 73.9 bar to 300 bar.

(19) In some embodiments, introducing solid materials into a first pressure vessel comprises mixing the solid materials with a liquid to form a slurry and introducing the slurry into the first pressure vessel. In some embodiments, allowing the working fluid to permeate into the solid materials comprises allowing the liquid to absorb at least some of the working fluid and allowing at least some of the working fluid to permeate into the solid materials. In some embodiments, allowing the working fluid to permeate into the solid materials comprises separating the solid materials from the slurry and allowing the working fluid to permeate into the separated solid materials. In some embodiments, the method comprises re-combining the separated solid materials and the liquid to reform the slurry before rapidly expanding the working fluid to thereby create fractured solid materials from the solid materials. In some embodiments, rapidly expanding the working fluid to thereby create fractured solid materials from the solid materials comprises expelling the slurry from the first pressure vessel. In some embodiments, the method comprises directing the slurry at a solid surface to cause further pulverization of the fractured solid materials.

(20) In some embodiments the method comprises warming the solid materials before introducing the solid materials into the first pressure vessel.

(21) In cases where water is used, e.g. to form a slurry, the water may be reused. In cases where the water is reused the water may be conditioned before reuse, for example, to remove particles of the solid materials and/or to remove dissolved substances from the water. The water conditioning may, for example, include settling and/or filtration. In some embodiments the water conditioning removes carbonic acid from the water.

(22) In some embodiments, the method comprises introducing filler elements into gaps between the solid materials in the first pressure vessel. The filler elements take up space and thereby reduce the amount of working fluid required to achieve a desired pressure within the first pressure vessel. The filler elements may, for example, comprise elements of a metal such as steel or another material that is robust and not susceptible to itself being comminuted in performance of the method. For example, the filler elements may comprise steel balls. After the solid materials have been comminuted the filler elements may be separated and reused.

(23) In some embodiments, reducing the pressure of the working fluid to the second pressure comprises opening a valve to release at least some of the working fluid through the valve into a second pressure vessel for recapture.

(24) In some embodiments in which the working fluid is captured and recycled for use in

processing more solid materials, the working fluid is conditioned before it is recycled. For example, the conditioning may comprise separating entrained particles of the solid materials (e.g. rock dust or rock particles) from the working fluid. The separating may, for example, comprise filtering, settling, centrifugation or combinations of these. In cases where the working fluid is captured as a gas the conditioning may include repressurizing the working fluid. The repressurizing may carry the working fluid through a phase change e.g. from a gas to a liquid or from a gas to a supercritical fluid. For example, where the working fluid comprises CO<sub>2</sub>, the conditioning may comprise one or more of: cooling, condensing and liquifying the CO<sub>2</sub> before increasing the pressure of the liquefied CO<sub>2</sub> to cause the CO<sub>2</sub> to undergo a phase change to supercritical CO<sub>2</sub>; directly compressing captured gaseous CO<sub>2</sub> to cause a transition to a supercritical phase and removing heat as required; a hybrid combination comprising both pumping liquefied CO<sub>2</sub> to a pressure sufficient to yield supercritical CO<sub>2</sub> and also compressing some gaseous CO<sub>2</sub> to yield more supercritical CO<sub>2</sub>.

(25) In some embodiments, after rapidly expanding the working fluid and before removing the fractured solid materials from the first pressure vessel, the method comprises increasing the pressure of the working fluid in the first pressure vessel to the operating pressure, and rapidly expanding the working fluid to further fracture the fractured solid materials.

(26) In some embodiments, after rapidly expanding the working fluid and before removing the fractured solid materials from the first pressure vessel, the method comprises increasing the temperature of the working fluid in the first pressure vessel to the operating temperature, and rapidly expanding the working fluid to further fracture the fractured solid materials.

(27) In some embodiments, after rapidly expanding the working fluid and before removing the fractured solid materials from the first pressure vessel, the method comprises increasing the temperature of the working fluid in the first pressure vessel to the operating temperature, increasing the pressure of the working fluid in the first pressure vessel to the operating pressure, and rapidly expanding the working fluid to further fracture the fractured solid materials.

(28) In some embodiments, increasing the pressure of the working fluid to the operating pressure comprises injecting working fluid recaptured in a second pressure vessel into the first pressure vessel, wherein the recaptured working fluid was recaptured during a step of rapidly expanding the working fluid.

(29) The solid materials may be introduced into the first pressure vessel either while the first pressure vessel is depressurized or while a pressure exceeding atmospheric pressure is maintained in the first pressure vessel. In some embodiments, introducing solid materials into the first pressure vessel comprises introducing solid materials into the first pressure vessel while maintaining a pressure within the first pressure vessel greater than atmospheric pressure.

(30) In some embodiments, introducing solid materials into the pressure vessel comprises introducing solid materials into the first pressure vessel through a first airlock, a first lock hopper or a first piston feed.

(31) In some embodiments, the method comprises removing the fractured solid materials from the pressure vessel while maintaining a pressure within the first pressure vessel greater than atmospheric pressure.

(32) In some embodiments, removing the fractured solid materials from the first pressure vessel comprises removing the fractured solid materials from the first pressure vessel through the first airlock, a second airlock, the first lock hopper, a second lock hopper, the first piston feed, a second piston feed, a rotary valve, a plug-forming feeder or a dynamic feeder.

(33) Some embodiments include purging air and/or non-condensing fluids from the system. For example the first pressure vessel may be purged before or during introducing the working fluid into the first pressure vessel. In some cases some of the working fluid may be intermixed with air or other fluids being purged. In such cases the method may include separating the intermixed working

fluid from the other fluids being purged (e.g. CO.sub.2 may be separated from purged air and/or water).

(34) Another aspect of the invention provides an apparatus for comminution of solid materials. The apparatus may comprise means for performing any of the steps of the methods described herein and the methods may include any mode of using the apparatus as described herein. The apparatus may comprise a first pressure vessel, the first pressure vessel comprising a first chamber, a first pump connected to pump a working fluid into the first chamber through a second inlet, a second pressure vessel connected to the first chamber by a first valve, wherein when the first valve is open, at least some of the working fluid within the first chamber is allowed to flow out of a first outlet and into a second chamber of the second pressure vessel.

(35) In some embodiments, the apparatus comprises a temperature sensor for measuring a temperature of contents of the first pressure vessel, a thermal control system for heating or cooling the contents of the first pressure vessel, and a temperature controller for controlling the thermal control system based at least in part on measurements from the temperature sensor.

(36) In some embodiments, the temperature controller is configured to maintain the temperature of the contents of the first pressure vessel at an operating temperature of the working fluid. In some embodiments, the temperature controller is configured to maintain the temperature of the contents of the first pressure vessel above a critical temperature of the working fluid.

(37) In some embodiments, the apparatus comprises a pressure sensor for determining a pressure inside the first chamber, and a pressure controller for controlling the first pump based at least in part on measurements from the pressure sensor.

(38) In some embodiments, the pressure controller is configured to maintain the pressure inside the first chamber at an operating pressure. In some embodiments, the pressure controller is configured to maintain the pressure inside the first chamber above a critical pressure. In some embodiments, the pressure controller is configured to maintain the pressure inside the first chamber above a critical pressure of the working fluid for less than 60 seconds before the first valve is opened.

(39) In some embodiments, the second pressure vessel is connected to the first pump and the first pump is connected to pump working fluid from the second pressure vessel into the first pressure vessel.

(40) In some embodiments, the apparatus comprises a second pump configured to pump working fluid from the second pressure vessel into the first pressure vessel.

(41) In some embodiments, the first inlet comprises a first airlock, a first lock hopper or a first piston feed.

(42) In some embodiments, the first pressure vessel comprises a second outlet for removing fractured solid materials from the first chamber, the second outlet comprising the first airlock, a second airlock, the first lock hopper, a second lock hopper, the first piston feed, a second piston feed, a rotary valve, a plug-forming feeder or a dynamic feeder.

(43) Further aspects and example embodiments are illustrated in the accompanying drawings, the claims and/or described in the following description.

(44) It is emphasized that the invention relates to all combinations of the above features, even if these are recited in different claims.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Exemplary embodiments are illustrated in referenced figures of the drawings. It is intended that the embodiments and figures disclosed herein are to be considered illustrative rather than restrictive.

(2) FIG. 1 depicts an exemplary non-limiting method for comminution of solid materials according

to one embodiment of the invention.

(3) FIG. 2 depicts an exemplary non-limiting apparatus for comminution of solid materials according to one embodiment of the invention.

(4) FIG. 3 depicts another exemplary non-limiting method for comminution of solid materials according to one embodiment of the invention.

(5) FIG. 4 depicts another exemplary non-limiting apparatus for comminution of solid materials according to one embodiment of the invention.

(6) FIG. 5 is a schematic illustration showing apparatus for comminution of solid materials according to another example embodiment of the invention.

#### DESCRIPTION

(7) Throughout the following description specific details are set forth in order to provide a more thorough understanding to persons skilled in the art. However, well known elements may not have been shown or described in detail to avoid unnecessarily obscuring the disclosure. Accordingly, the description and drawings are to be regarded in an illustrative, rather than a restrictive, sense.

(8) One aspect of the invention provides methods for comminution of solid materials. In some embodiments, the solid materials comprise ore-bearing rocks. In some embodiments, the solid materials comprise non-ore-bearing rocks such as limestone. The method(s) may comprise introducing solid materials into a first pressure vessel and injecting a working fluid, such as CO<sub>2</sub> into the first pressure vessel. The pressure and temperature inside of the first pressure vessel may be maintained at a desired pressure and a desired temperature. In some embodiments, the working fluid (e.g. CO<sub>2</sub>) is in a supercritical phase at the desired pressure and temperature. In some embodiments, the supercritical phase is a liquid or vapour phase. In some embodiments, the working fluid is a “dense fluid” (e.g. when pressure is above the critical pressure of the working fluid and the temperature is below the critical temperature of the working fluid) at the desired pressure and temperature. The working fluid may be allowed to penetrate and/or permeate into the solid materials (e.g. through one or more pores of the solid materials). Once sufficient permeation is achieved, the working fluid may then be caused to expand (e.g. decrease in density) at a high rate. The rapid expansion may be achieved by causing a phase change (e.g. from a supercritical phase to a subcritical vapour phase or from a liquid phase to a vapour phase, etc.). The phase change of the working fluid may be achieved by varying the pressure and/or the temperature of the working fluid within the first pressure vessel. When the working fluid expands, it may drive a pressure wave through the solid materials from within cracks and pores of solid materials, thereby causing rock fracturing and/or pulverization driven from within. In some embodiments, the temperature and/or pressure and/or phase of the working fluid is varied repeatedly to cause a plurality of rapid expansions of the working fluid. Each successive rapid expansion of the working fluid may cause further comminution of the solid materials until a desired particle size is achieved.

(9) FIG. 1 depicts an exemplary, non-limiting method **100** for comminution of solid materials **12** to produce fractured solid materials **12'**. FIG. 2 depicts an exemplary, non-limiting apparatus **10** that may be employed to carry out method **100**. It should be understood that method **100** may be carried out on apparatuses other than apparatus **10** and methods other than method **100** may be carried out on apparatus **10**.

(10) Solid materials **12** may comprise porous solid materials. For example, solid materials **12** may comprise rocks. In some embodiments, solid materials **12** comprise ore-bearing rocks. The ore may comprise, without limitation, oxides, sulfides, silicates, native metals such as nickel, zinc, copper, noble metals such as gold, platinum or silver, etc. In some embodiments, solid materials **12** comprise non-ore-bearing rocks such as, but not limited to, limestone.

(11) Solid materials **12** may vary in size. For example, in some embodiments, solid materials **12** comprise particles having a maximum dimension of between approximately 5 cm and 3 m or between approximately 50 cm and 2 m. In some embodiments, solid materials **12** comprise particles having a maximum dimension of less than approximately 5 cm or less than 2.5 cm. In

some embodiments, solid materials **12** comprise a mixture of particles of different sizes. In some embodiments, solid materials **12** can be sorted prior to performing method **100**.

(12) Step **102** of method **100** comprises introducing solid materials **12** into a first chamber **14A** of a first pressure vessel **14**. In some embodiments, solid materials **12** are dried, wetted, and/or cleaned prior to step **102**, although this is not mandatory. First pressure vessel **14** may comprise any suitable pressure vessel.

(13) In some embodiments, introducing solid materials **12** into first chamber **14A** comprises depressurizing first chamber **14A** (e.g. to at or near atmospheric pressure) before inserting solid materials **12** into first chamber **14A**. For example, in the illustrated embodiment, first chamber **14A** may be depressurized before a cover **14D** of first pressure vessel **14** is removed to provide access to first chamber **14A** to facilitate insertion of solid materials **12**. In some embodiments, after solid materials **12** are introduced into first chamber **14A** of first pressure vessel **14**, first pressure vessel **14** may be sealed (e.g. by fastening cover **14D**).

(14) In some embodiments, first pressure vessel **14** is configured to allow introduction of solid materials **12** into first chamber **14A** without depressurizing first chamber **14A** or while maintaining a pressure greater than atmospheric pressure within first chamber **14A**. For example, an airlock, rotary valves, a lock hopper, plug-forming feeders, piston feeders, dynamic feeders, slurry feeders or the like may be provided for insertion of solid materials **12** into first chamber **14A** while maintaining first chamber **14A** at a pressure greater than atmospheric pressure.

(15) For example, in some embodiments, a lock hopper (not depicted) is provided to facilitate introduction of solid materials **12** into first chamber **14A** without depressurizing first chamber **14A** or while maintaining a pressure greater than atmospheric pressure within first chamber **14A**. The lock hopper may comprise an input valve from atmosphere to the lock hopper chamber, and an output valve from the lock hopper chamber to the first chamber **14A**. With the output valve closed, the input valve is opened to allow delivery of solid materials **12** into the lock hopper at atmospheric pressure. After solid materials **12** are in the lock hopper chamber, the input valve is closed. The pressure inside the lock hopper chamber is then increased to approximately the same pressure as is present in first chamber **14A**. This may be accomplished by introducing fluid from first chamber **14A** into the lock hopper chamber and/or by additional pumping or compression apparatus such as a pump, compressor or pressurized chamber. The output valve of the lock hopper is subsequently opened to allow solid materials **12** to flow into the first chamber **14A**. The output valve is then closed, and the lock hopper chamber pressure is reduced. This may be accomplished by returning fluid back to first chamber **14A**, or otherwise.

(16) As another example, in some embodiments, a piston feeder (not depicted) is provided to allow introduction of solid materials **12** into first chamber **14A** without depressurizing first chamber **14A** or while maintaining a pressure greater than atmospheric pressure within first chamber **14A**. A piston feeder may function in a similar manner to the lock hopper described above but may employ ports instead of valves. The ports may be opened or closed by moving the piston from a first position (in which an input port is open and an output port is closed) to a second position (in which the input port is closed and the output port is open).

(17) Step **104** of method **100** comprises providing a working fluid **16** in first pressure vessel **14**. In some embodiments, working fluid **16** is present in first pressure vessel before step **104**. In some embodiments, air is removed from first pressure vessel **14** prior to or during insertion of working fluid **16** into first pressure vessel **14**, although this is not mandatory. For example, in some embodiments, a vacuum removes air from first pressure vessel **14** and/or air is purged from first pressure vessel **14** by insertion of working fluid **16** or another suitable fluid (e.g. an inert gas).

(18) Working fluid **16** may comprise any suitable working fluid. In some embodiments, working fluid **16** comprises a fluid having a critical pressure in the range of about 73 bar to about 221 bar and a critical temperature in the range of about 31° C. to about 374° C. In some embodiments, working fluid **16** has low surface tension low viscosity, high diffusivity and/or high solubility. In



some embodiments, working fluid **16** comprises water. In some embodiments, working fluid **16** comprises nitrogen gas. In some embodiments, working fluid **16** comprises ethane. In some embodiments, working fluid **16** comprises CO.sub.2.

(19) In some embodiments, working fluid **16** is delivered, for example, from a working fluid source **16A** such as, a gas cylinder and through a conduit **16B** to first pressure vessel **14**. In some embodiments, recaptured working fluid **16** is delivered to first pressure vessel **14** from a second pressure vessel **28**, as is discussed further herein.

(20) Step **104** comprises providing working fluid **16** in first chamber **14A** at an operating pressure, P.sub.op, and an operating temperature, T.sub.op. In some embodiments, working fluid **16** is injected into first chamber **14A** at the operating pressure, P.sub.op, and/or the operating temperature, T.sub.op. In other embodiments, step **104** comprises raising pressure and/or temperature of working fluid **16** to achieve the operating pressure, P.sub.op, and the operating temperature, T.sub.op, after working fluid **16** is injected into first chamber **14A**.

(21) In some embodiments, the operating pressure, P.sub.op, and the operating temperature, T.sub.op, are chosen to achieve a supercritical phase of working fluid **16**. In some embodiments, the operating pressure, P.sub.op, comprises a pressure greater than the critical pressure of working fluid **16** and the operating temperature, T.sub.op, comprises a temperature that is greater than the critical temperature of working fluid **16**. In some embodiments, the operating pressure, P.sub.op, and the operating temperature, T.sub.op, are chosen to achieve a dense fluid phase of working fluid **16** (e.g. the operating pressure, P.sub.op, is above the critical pressure of working fluid **16** and the operating temperature, T.sub.op, is below the critical temperature of working fluid **16**).

(22) For example, CO.sub.2 is in its supercritical phase at a pressure of greater than 73.8 bar and a temperature of greater than 31.0° C. As such, where working fluid **16** comprises CO.sub.2, the operating pressure, P.sub.op, comprises a pressure of greater than 73.8 bar and the operating temperature, T.sub.op, comprises a temperature of greater than 31.0° C. (to achieve supercritical CO.sub.2) or the operating temperature, T.sub.op, comprises a temperature of below 31.0° C. (to achieve dense fluid CO.sub.2). In some embodiments where working fluid **16** comprises CO.sub.2, the operating pressure, P.sub.op, comprises a pressure of greater than 73.8 bar and less than approximately 200 bar or 300 bar. In some embodiments, the operating temperature, T.sub.op, comprises a temperature of greater than 31.0° C. and less than approximately 100.0° C.

(23) CO.sub.2 is a desirable working fluid **16** for a variety of reasons. The energy efficiency of method **100** may be improved since CO.sub.2 is in its supercritical phase at relatively low temperatures (e.g. temperature of greater than 31.0° C.). Similarly, the energy efficiency of method **100** may be improved since CO.sub.2 may be in its supercritical phase at relatively low pressures (e.g. pressures greater than 73.8 bar). Further, CO.sub.2 is easily obtainable, inexpensive, has low human hazard and has relatively low environmental impact. In some embodiments, CO.sub.2 working fluid **16** may be obtained through direct capture from air, flue gas, exhaust gases, industrial gases or the like. CO.sub.2 may be obtained using a commercially available carbon capture system. Further, CO.sub.2 has minimal interaction with most ore-bearing rocks and will therefore not substantially degrade the resultant products of method **100**.

(24) In some embodiments, a pump (or compressor) **18** is provided to achieve a desired pressure of working fluid **16** within first chamber **14A** of first pressure vessel **14** at step **104**. Pump **18** may comprise any suitable pump or compressor.

(25) In some embodiments, working fluid **16** in a gas or vapour phase is chilled and condensed into a liquid phase at an intermediate pressure and/or temperature (e.g. above approximately 5.2 bar and between approximately -56.6° C. and +31.1° C.) and the liquid phase working fluid **16** is pumped into first chamber **14A** at step **104**.

(26) In other embodiments, the pressure of working fluid **16** is increased at step **104** by decreasing a volume of first chamber **14A**. For example, a sealed piston may be employed to reduce a volume of first chamber **14A** or a non-volatile liquid may be pumped into first chamber **14A** to reduce a

volume of first chamber **14A**.

(27) In some embodiments, the pressure of working fluid **16** is increased at step **104** by employing combustion. A first volume of working fluid **16** may be delivered into first chamber **14A**. A mixture of air (and/or oxygen) and fuel (e.g. propane, diesel, natural gas, syngas, etc.) is introduced into a second chamber (e.g. second chamber **28A** or another chamber, not depicted) connected to first chamber **14A**. The mixture in the second chamber is then ignited thereby causing a rapid increase in pressure in the second chamber. In some embodiments, at least some of the fluid generated by ignition of the mixture in the second chamber may be delivered to first chamber **14A** to increase the pressure at step **104**. In other embodiments, the rapid expansion of fluid caused by the combustion may be employed to drive a piston (or the like) to reduce the volume of first chamber **14A** or otherwise compress working fluid **16**. Employing combustion at step **104** to increase the pressure of working fluid **16** may avoid losses that would otherwise be generated due to friction, inefficient mechanical pumping means, compressors, etc. and/or reduce or eliminate one or more moving parts that would otherwise be used for method **100** (e.g. compressors, fast acting valves, pressure regulators, pumps etc.). If concentrated oxygen is employed instead of air, the emissions of such combustion may be limited to (or substantially limited to) CO<sub>2</sub> (which may be used as working fluid **16**) and water. Moreover, the heat produced by the combustion may be employed to achieve T<sub>sub.op</sub> at step **104** of method **100**.

(28) A temperature of working fluid **16** may be raised or maintained within first pressure vessel **14** by employing a heat source **24**. Heat source **24** may comprise any suitable heat source. As non-limiting examples, heat source **24** may comprise a heating jacket wrapped around at least a portion of first pressure vessel **14**, an immersion heater placed within first pressure vessel **14**, and/or a heat exchanger arranged to transfer heat into working fluid supply being delivered to pressure vessel **14** (e.g. in a supply pipe, **16B**).

(29) Undesirable accumulation of heat or loss of heat may be reduced by one or more of insulation (e.g. around first pressure vessel **14**) and/or heat exchangers. For example, a heat exchanger could be provided to transfer heat removed from working fluid **16** at one step of process **100** (e.g. if working fluid **16** is cooled to achieve its liquid phase) to another step of process **100** (e.g. if working fluid **16** is heated to achieve its supercritical phase).

(30) Apparatus **10** may comprise one or more sensors and one or more controllers to monitor and/or control the temperature, pressure and/or density of working fluid **16** within first chamber **14A**. For example, in the illustrated embodiment, apparatus **10** comprises a pressure sensor **20A** for measuring the pressure within first chamber **14A**, a temperature sensor **20B** for measuring the temperature within first chamber **14A** and a density sensor **20C** for measuring density of working fluid within first chamber **14A**. Pressure sensor **20A** and temperature sensor **20B** and density sensor **20C** are connected to a controller **22**. Controller **22** may monitor the pressure within first chamber **14A** and control pump **18** to achieve operating pressure, P<sub>sub.op</sub>. Similarly, controller **22** may monitor the temperature within first chamber **14A** and control a heat source **24** to achieve operating temperature, T<sub>sub.op</sub>. Similarly, controller **22** may monitor density of the working fluid within first chamber **14A** and control one or a combination of pump **18** and heat source **24** to achieve a desired operating density D<sub>sub.op</sub>.

(31) Apparatus **10** may comprise other sensors such as one or more mass flow sensors that monitor amounts of one or more materials of different kinds that are moved through apparatus **10**. For example, mass/mass flow sensors may be provided to monitor amounts of solids (e.g. rock) delivered into first chamber **14A**, amounts of working fluid **16** delivered into or recovered from first chamber **14A** and/or amounts of denser fluids such as liquid water delivered to or recovered from first chamber **14A**.

(32) Controller **22** may monitor the operational status of system **10** by way of sensors such as those described above and other sensors. As with other industrial controls, controller **22** may include an alarm system that triggers alarms if process parameters are out of limits and/or a malfunction of

apparatus **10** is likely.

(33) In some embodiments, one or more filters are provided to remove particulates (e.g. dust or dirt) from working fluid **16**. For example, cyclonic or spiral tube separation, or a rotating disk filtration system for solids separation from liquid CO<sub>2</sub> in a continuous pressure filtration system, (also known as hyperbaric filtration) may be employed to remove particulates (e.g. dust or dirt) from working fluid **16**.

(34) Step **106** of method **100** comprises allowing working fluid **16** to permeate into solid materials **12** during a permeation time,  $t_{sub.p}$ . In some embodiments, working fluid **16** permeates (or begins to permeate) into solid materials **12** during step **104** (e.g. during the time it takes to achieve operating pressure,  $P_{sub.op}$ , and/or operating temperature,  $T_{sub.op}$ ). In some embodiments, working fluid **16** permeates (or begins to permeate) into solid materials **12** in a first phase (e.g. vapour phase) and then working fluid **16** undergoes a phase change (e.g. to liquid or dense fluid) during step **104** or **106**. In some embodiments, permeation time,  $t_{sub.p}$ , is in the range of about 0 to about 60 seconds. In some embodiments, permeation time,  $t_{sub.p}$ , is in the range of about 0 to about 60 minutes. During step **106**, working fluid **16** may penetrate into a porous structure of solid materials **12**. In some embodiments, longer permeation times,  $t_{sub.p}$ , may increase the depth of permeation of working fluid **16** into solid materials **12** and may decrease the size of fractured solid materials **12'** resultant from method **100**. In some embodiments, it is desirable to maintain the operating pressure,  $P_{sub.op}$ , and the operating temperature,  $T_{sub.op}$  during  $t_{sub.p}$ . Controller **22** may be employed to maintain the operating pressure,  $P_{sub.op}$ , and the operating temperature,  $T_{sub.op}$  during  $t_{sub.p}$ .

(35) Step **108** comprises rapidly expanding working fluid **16**. For example, in some embodiments, the working fluid **16** is caused to expand by 2 times, 10 times, 100 times, 250 times or even more than 500 times. As working fluid **16** expands, working fluid **16** may drive a pressure wave through solid materials **12** from within cracks and pores of solid materials **12**, thereby causing fracturing and pulverization driven from within.

(36) In some embodiments, an excess volume of working fluid **16** produced at step **108** is vented. In some embodiments, the excess volume of working fluid **16** produced at step **108** is vented and recaptured in secondary pressure vessel **28**.

(37) For some solid materials **12** (e.g. rocks) the tensile strength of such solid materials **12** is much lower than the compressive strength. For this reason, method **100**, which takes advantage of the relatively low tensile strength of some solid materials **12**, may use substantially less energy than conventional crushing and grinding comminution methods that rely on compressive and/or shear forces to break solid materials (e.g. ore-bearing rocks).

(38) Various techniques may be employed to reduce the density of (i.e. expand) working fluid **16** at step **108**. In some embodiments, the density of working fluid **16** may be reduced by causing a phase change of working fluid **16**. For example, the density of working fluid **16** may be reduced by causing a phase change of working fluid **16** from a supercritical phase to a subcritical vapour phase. This phase change may cause a rapid expansion (e.g. at the speed of sound, or in special cases such as the unstable vaporization that may occur from dropping the working fluid pressure into the vapour dome and crossing the spinodal curve from the superheated liquid side, a supersonic expansion) of working fluid **16**.

(39) The phase change may be achieved by lowering the pressure of working fluid **16** within first chamber **14A**. The phase change may be achieved by lowering of the pressure to below a critical pressure of working fluid **16**. For example, where working fluid **16** comprises CO<sub>2</sub>, the working fluid **16** may be expanded by reducing the pressure of working fluid **16** within first chamber **14A** to below 73.8 bar.

(40) In some embodiments, step **108** comprises reducing the pressure of working fluid **16** within first chamber **14A** by releasing a mass of working fluid **16** through an outlet **14B**. Outlet **14B** may be opened and closed by one or more valves **26**. Valve(s) **26** may be controlled manually by an

operator or by a controller (e.g. controller **22**). Valve(s) **26** may comprise any suitable valve(s). For example, valve(s) **26** may comprise a snap open or pop open valve. Valve(s) **26** may be capable of fully opening in approximately 30 ms, 10 ms, 5 ms, 3 ms, 1 ms or less.

(41) Outlet **14B** may comprise one or more filters, particle separators, particle trapping processors or the like to prevent particles of solid materials **12** from escaping first chamber **14A** undesirably through outlet **14B**. Valve(s) **26** may comprise one or more features to prevent particles of solid materials **12** from escaping first chamber **14A** undesirably.

(42) In some embodiments, as a pressure of working fluid **16** within first chamber **14A** is increased, some working fluid **16** within first chamber **14A** is exhausted to prevent damage to first pressure vessel **14**. In some embodiments, working fluid **16** is exhausted through one or more blow-off valves configured to prevent the pressure in first chamber **14A** from exceeding a threshold. In some embodiments, such blow-off valves (e.g. blow-off valves **14C**) are separate from outlet **14B**. In some embodiments, such blow-off valves are integrated with outlet **14B** to allow any exhausted working fluid **16** to be recaptured with other working fluid **16** exhausted through outlet **14B**, if desired.

(43) In some embodiments, method **100** comprises repeating steps **104** through **108** one or more times for the same instance of solid materials **12**. Each time that steps **104** through **108** are repeated, working fluid **16** may be able to penetrate deeper into the porous structure of solid materials **12** which may allow for further reduction of size of solid materials **12**. In some embodiments, solid materials **12** are screened for size after removal from first chamber **14A**. Any particles that are not sufficiently small may be returned to first chamber **14A** for further processing. In some embodiments, solid materials **12** are screened for size before removal from first chamber **14A**. In this way, solid materials **12** may undergo repeated cycles of steps **104** to **108** until sufficient comminution has occurred without being removed from first chamber **14A**.

(44) In some embodiments apparatus **10** provides a way to monitor a degree of comminution of solid materials in first pressure vessel **14**. For example, first pressure vessel **14** may comprise a window or a camera to allow an operator to see into first chamber **14A** to determine if sufficient comminution has occurred and/or apparatus **10** may include a remote sensing system such as an ultrasound imaging system or an acoustic sensor that produces output images or signals that are indicative of a degree of comminution of the solid materials in pressure vessel **14A**. It may be possible to predict a desired number of cycles of steps **104** through **110** based on one or more characteristics (e.g. mass, composition, volume, porosity, etc.) of solid materials **12**.

(45) In some embodiments, working fluid **16** that is exhausted through outlet **14B** is exhausted into the atmosphere. In some embodiments, working fluid **16** that is exhausted through outlet **14B** is recaptured for further use.

(46) In some embodiments, working fluid **16** that is exhausted through outlet **14B** is recaptured in a second chamber **28A** of a second pressure vessel **28**. Second pressure vessel **28** may comprise any suitable pressure vessel. In some embodiments, second chamber **28A** has a greater volume than first chamber **14A** to accommodate lower density working fluid **16** exhausted from outlet **14B**.

(47) In some embodiments, working fluid **16** recaptured in second chamber **28A** has a pressure greater than atmospheric pressure. Therefore, to increase the pressure of working fluid **16** recaptured in second chamber **28A** to operating pressure,  $P_{sub.op}$ , involves a smaller increase in pressure (than if pressurized from atmospheric pressure) and less energy to do so. Further, by recapturing working fluid **16** instead of venting it into the atmosphere, less working fluid **16** is released into the atmosphere and potential environmental harm of method **100** is reduced.

(48) In some embodiments, working fluid **16** recaptured in second chamber **28A** is injected into first chamber **14A** at step **104** to achieve  $T_{sub.op}$  and  $P_{sub.op}$ .

(49) In some embodiments, working fluid **16** flows from second chamber **28A** to first chamber **14A** at step **104** through conduit **16B** and pump **18**. Optionally, pump **18** pressurizes working fluid in a high pressure reservoir **23** and the pressurized working fluid **16** is metered into first chamber **14A**.

by way of a controlled valve 23A.

(50) In some example embodiments, a valve 32 is employed to control whether working fluid 16 is delivered to first chamber 14A from working fluid source 16A, from second pressure vessel 28A or some combination thereof. Valve 32 may be controlled manually or by a controller (e.g. controller 22). For example, if controller 22 (or an operator) determines that there is an insufficient pressure of working fluid 16 within second pressure vessel 28, working fluid 16 may be delivered from working fluid source 16A at step 104.

(51) In some embodiments, working fluid 16 flows from second chamber 28A to first chamber 14A at step 104 through valve(s) 26 and outlet 14B. An optional pump or compressor (not illustrated) may be provided to deliver working fluid 16 into first chamber 14A through outlet 14B. In some embodiments, the optional pump or compressor is integrated into second pressure vessel 28.

(52) In some embodiments, working fluid 16 is provided to second pressure vessel 28 from working fluid source 16A to counteract undesirable loss of working fluid 16 or to increase a ratio of working fluid 16 to other fluids within second pressure vessel 28.

(53) Apparatus 10 may comprise one or more sensors and one or more controllers to monitor and/or control the temperature and pressure of working fluid 16 within second chamber 28A. For example, in the illustrated embodiment, apparatus 10 comprises a pressure sensor 30A for measuring the pressure within second chamber 28A and a temperature sensor 30B for measuring the temperature within second chamber 28A. Pressure sensor 30A and temperature sensor 30B may be connected to controller 22 (or another suitable controller). Controller 22 may monitor the pressure within second chamber 28A to facilitate transfer of working fluid 16 from second chamber 28A to first chamber 14A. Similarly, controller 22 may monitor the temperature within second chamber 28A to facilitate transfer of working fluid 16 from second chamber 28A to first chamber 14A.

(54) In some embodiments, working fluid 16 is exhausted through one or more blow-off valves 28B configured to prevent the pressure in second chamber 28A from exceeding a threshold value.

(55) After step 108, if solid materials 12 are sufficiently pulverized (or if steps 104 to 108 have been repeated a designated number of times), method 100 continues to step 110. At step 110, fractured solid materials 12' are removed from first pressure vessel 14.

(56) In some embodiments, removing fractured solid materials 12' from first chamber 14A comprises depressurizing first chamber 14A (e.g. to at or near atmospheric pressure) before removing fractured solid materials 12' from first chamber 14A by, for example, removing cover 14D.

(57) In other embodiments, fractured solid materials 12' are removed without depressurizing first chamber 14A or while maintaining a pressure in first chamber 14A that is higher than atmospheric pressure. For example, an airlock, rotary valves, a lock hopper, plug-forming feeders, piston feeders, dynamic feeders and/or slurry feeders may be employed for removal of fractured solid materials 12' while first chamber 14A is maintained at a pressure greater than atmospheric pressure. In some embodiments, the same feature (e.g. an airlock, rotary valves, a lock hopper, plug-forming feeders, piston feeders, dynamic feeders and/or slurry feeders) is used for insertion and removal of solid materials 12 and fractured solid materials 12'. In some embodiments, separate features (e.g. an airlock, rotary valves, a lock hopper, plug-forming feeders, piston feeders, dynamic feeders and/or slurry feeders) are provided for insertion and removal of solid materials 12 and fractured solid materials 12'.

(58) In some embodiments solid materials 12 and working fluid 16 are ejected from first chamber 14A into second chamber 28A together. For example, upon opening one or more valves, the contents of first chamber 14A (e.g. solid material 12, working fluid, filler material if present, and slurrying liquid, if present) may be ejected from first chamber 14A into second chamber 28A. The ejection may be driven by pressure differential between the higher pressure of first chamber 14A and the significantly lower pressure of second chamber 28A. In some embodiments first chamber

**14A** is located above second chamber **28A** so that the ejection is assisted by gravity.

(59) Upon opening the valve(s) to allow the ejection of the materials from first chamber **14A** to second chamber **28A**, the materials experience a sudden pressure drop. This pressure drop may cause the working fluid to transition from a supercritical phase to a gas phase.

(60) The rate at which the pressure drops may be increased by making the valve(s) through which the materials are ejected have a large cross sectional area, using plural valves, and/or opening the valves rapidly. In some embodiments the materials exit into second chamber **28A** through a nozzle shaped to provide a rapid depressurization of materials as they pass out of the nozzle into second chamber **28A**.

(61) The magnitude of the pressure drop depends at least in part on the relative volumes of first chamber **14A** and second chamber **28A**. The magnitude of the pressure drop may be increased by increasing the size of second chamber **28A**. In some embodiments the reduced pressure after the pressure drop remains significantly higher than atmospheric pressure.

(62) In some embodiments the mixture of solid material **12** and working fluid is depressurized in first chamber **14A** and subsequently released into second chamber **28A** for collection/separation etc. Such embodiments may include valves that may be opened to rapidly vent working fluid from first chamber **14A** to cause a sudden pressure drop in first chamber **14A**. For example, pressure may be equalized or partially equalized between first chamber **14A** and second chamber **28A** and then fractured solid material **12** may be transferred to second chamber **28A** by opening a gate, chute, or the like.

(63) In some embodiments the mixture of solid material **12** and working fluid in first chamber **14A** is cycled two or more times between a higher pressure and a lower pressure. The profile of pressure as a function of time may include rapid decompressions. The rapid decompressions may cause the working fluid in first chamber **14A** to transition from a supercritical state to a gas state, a liquid state or a multiphase gas/liquid state. Between the rapid decompressions the pressure may be increased so that working fluid in first chamber **14A** is again in the supercritical state. Subsequently the mixture of solid material **12** and working fluid in first chamber **14A** may be ejected into second chamber **28A** as described above.

(64) In some embodiments first chamber **14A** includes a series of two or more chambers. Solid material **12** may be processed in sequence in the two or more chambers. In each of the chambers, the solid material may be received, pressurized with working fluid (optionally pressurized and depressurized in the chamber one or more times), and subsequently expelled into a next one of the chambers (or expelled into second chamber **28A** from the last one of the two or more first chambers **14A**).

(65) In any of the described embodiments, working fluid may be collected and reused. For example, working fluid may be collected from second chamber **28A** and/or from another volume into which the working fluid has been vented for reuse.

(66) In some embodiments fractured material **12'** has a particle size of 1 mm or less or 500  $\mu\text{m}$  or less or 250  $\mu\text{m}$  or less. The apparatus and methods described herein may, for example, be applied in applications for which ball mills are conventionally used.

(67) FIG. 3 depicts another exemplary, non-limiting method **200** for comminution of solid materials **12** to produce fractured solid materials **12'**. FIG. 4 depicts an exemplary, non-limiting apparatus **300** that may be employed to carry out method **200**. It should be understood that method **200** may be carried out on apparatuses other than apparatus **300** and methods other than method **200** may be carried out on apparatus **300**.

(68) Method **200** is substantially similar to method **100** except as described herein. For example, like method **10**, method **200** involves allowing a working fluid **16** to permeate solid materials **12** and then rapidly expanding working fluid **16** to produce fractured solid materials **12'**. However, unlike method **10**, method **200** comprises mixing solid materials **12** with a liquid **34** to form a slurry **36** for use in part of, or throughout method **200**, as described further herein. Likewise,

apparatus **300** is substantially similar to apparatus **100** except as described herein. For example, like apparatus **100**, apparatus **300** comprises a first pressure vessel **314** having a first pressure chamber **314A** in which solid materials **12** (as part of slurry **36**) are exposed to working fluid **16** at operating pressure, P.sub.op, and operating temperature, T.sub.op. However, unlike in apparatus **10**, the pressure of working fluid **16** is maintained at a relatively constant level throughout method **200** and working fluid **16** that has permeated into solid materials **12** (as part of slurry **36**) is caused to rapidly expand by expelling slurry **36** from first pressure chamber **314**, as described further herein.

(69) Method **200** may start at step **202**. Step **202** comprises preparing a slurry **36**. Slurry **36** may comprise a mixture of solid materials **12** and liquid **34**. In some embodiments, liquid **34** comprises water. In some embodiments, liquid **34** comprises liquid CO.sub.2. In some embodiments, liquid **34** comprises water with some dissolved salt. The dissolved salt may reduce the ability of liquid **34** to absorb working fluid **16**, which may reduce the amount of working fluid **16** required at step **206**. In some embodiments liquid **34** includes one or more leaching agents. The leaching agents may, for example include acids. The leaching agents may help to liberate desired minerals from solid material **12** (e.g. by leaching) while the solid material is being processed in method **200**.

(70) Slurry **36** may be prepared by any suitable technique. In some embodiments, slurry **36** is made by mixing solid materials **12** and liquid **34** in a mixing tank such as, but not limited to, a rotating drum. In some embodiments, solid materials **12** may be sufficiently fine that combining solid materials **12** and liquid **34** in a tank (or the like) may provide sufficient mixing. In some embodiments, slurry **36** may be processed in one or more ways before method **200** proceeds to step **204**. For example, slurry **36** may be aerated, filtered, heated, etc. In some embodiments, working fluid **16** may be dissolved in slurry **36** before proceeding to step **204**.

(71) At step **204**, slurry **36** is introduced into a first chamber **314A** of a first pressure vessel **314**. Slurry **36** may be introduced into first pressure vessel **314** through an inlet **314B** by one or more pumps **318**. Pump(s) **318** may comprise one or more centrifugal pumps, piston pumps, rotary vane pumps, etc. Pump(s) **318** may allow for continuous delivery of slurry **36** into first chamber **314A** of first pressure vessel **314**. A rate of flow of slurry **36** through inlet **314B** may be monitored using one or more flow sensors. Pump(s) **318** may be controlled based at least in part on the output of such flow sensors.

(72) At step **206**, a working fluid **16** is provided in first pressure vessel **314** at operating temperature, T.sub.op and operating pressure, P.sub.op. Step **206** may be substantially similar to step **104** except that step **206** may occur continuously during method **200**. Working fluid **16** may comprise any suitable working fluid, as discussed herein. As compared to step **104**, step **206** may use a relatively lower volume of working fluid **16** due to the amount of space inside first chamber **314A** occupied by fluid **34** of slurry **36**. Moreover, because it uses less energy to pressurize an equal volume of liquid and an equal volume of gas, method **200** may use less energy to achieve P.sub.op as compared to method **100**.

(73) Like apparatus **10**, apparatus **300** may comprise one or more sensors and one or more controllers to monitor and/or control the temperature and pressure of working fluid **16** and slurry **36** within first chamber **314A**. For example, in the illustrated embodiment, apparatus **300** comprises a pressure sensor **320A** for measuring the pressure within first chamber **314A** and a temperature sensor **320B** for measuring the temperature within first chamber **314A**. Pressure sensor **320A** and temperature sensor **320B** are connected to a controller **322**. Controller **322** may monitor the pressure within first chamber **314A** and controller **322** may control pump **318**, a valve **314D** of outlet **314C** and/or a valve **332** of a working fluid source **316A** to achieve operating pressure, P.sub.op. Similarly, controller **322** may monitor the temperature within first chamber **314A** and control a heat source **324** (which may be similar to heat source **24** of apparatus **10**) to achieve operating temperature, T.sub.op.

(74) Step **208** of method **200** comprises allowing working fluid **16** to permeate into solid materials **12** within slurry **36** during a permeation time, t.sub.p to form a permeated slurry **36'**. Step **208** may

be substantially similar to step **106** of method **100**, except for the presence of liquid **34**. In some embodiments, permeation time,  $t_{sub.p}$ , is in the range of about 0 to about 60 seconds. During step **106**, working fluid **16** may penetrate into a porous structure of solid materials **12**. In some embodiments, longer permeation times,  $t_{sub.p}$ , may increase the depth of permeation of working fluid **16** into solid materials **12** and may decrease the size of fractured solid materials **12'** resultant from method **200**.

(75) At step **208**, some working fluid **16** may be absorbed by or dissolved in liquid **34** of slurry **36**. In some embodiments, liquid **34** is caused to be saturated with working fluid **16** such that additional working fluid **16** can better permeate solid materials **12**. Due to surface tension, working fluid **16** may permeate solid materials **12** while liquid **34** does not substantially permeate solid materials **12**.

(76) In some embodiments, to improve permeation of working fluid **16** into solid materials **12**, step **208** of method **200** comprises separating slurry **36** back into solid materials **12** and liquid **34** within first pressure chamber **314A**. This may be accomplished by one or more screens provided within first pressure chamber **314A**, a cyclonic particle separator, a screw conveyor with a screen or any other suitable apparatus. Once separated, solid materials **12** may be permeated by working fluid **16** without or with minimized interference by liquid **34** (e.g. similar to step **106** of method **100**). After solid materials **12** are sufficiently permeated by working fluid **16**, solid materials **12** may optionally be re-combined with liquid **34** to form a permeated slurry **36'**.

(77) Once solid materials **12** are sufficiently permeated by working fluid **16**, permeated slurry **36'** is expelled from first pressure chamber **314A** through outlet **314C**. In some embodiments, outlet **314C** comprises an orifice, nozzle or flow restrictor sized to allow a measured amount of permeated slurry **36'** to be expelled at a given pressure. In some embodiments, the rate of expulsion of permeated slurry **36'** from first chamber **314A** is maintained substantially equal to the rate of introduction of slurry **36** into first chamber **314A**. Maintaining these rates approximately equal to one another, may facilitate maintenance of  $P_{sub.op}$  within first chamber **314A** during method **200**.

(78) As solid materials **12** are expelled from outlet **314C**, solid materials **12** will experience a rapid and significant drop in pressure (e.g. from  $P_{sub.op}$  to atmospheric pressure or to a reduced pressure that is between  $P_{sub.op}$  and atmospheric pressure). This rapid and significant drop in pressure may cause working fluid **16** within solid materials **12** to rapidly expand (e.g. similar to in step **108** of method **100**) at step **212**. As working fluid **16** expands, working fluid **16** may drive a pressure wave through solid materials **12** from within cracks and pores of solid materials **12**, thereby causing fracturing and pulverization driven from within to produce fractured solid materials **12'**.

(79) Unlike method **100** and apparatus **10**, method **200** may not need a rapid expansion valve (e.g. valve **26**) or other mechanical means to cause a rapid decrease in pressure. As such, method **200** and apparatus **300** may be less costly and easier to maintain. Further, by not relying on a change in pressure within first pressure vessel **314**, method **200** and apparatus **300** do not use additional energy for repeated increases in pressure within first chamber **314A**. Further still, by not relying on a change in pressure within first pressure vessel **314**, method **200** and apparatus **300** may be operated with a continuous feed of slurry **36**.

(80) In some embodiments, outlet **314C** may direct permeated slurry **36'** toward an interaction zone in which impact of fractured solids **12'** against other objects in the interaction zone may cause further comminution of solid materials **12** within permeated slurry **36'**. The interaction zone may, for example comprise a hard surface, a pool of permeated slurry **36'** or other expelled fractured solid materials **12'**. Since permeated slurry **36'** may be expelled from outlet **314C** at a high velocity, the impact of permeated slurry **36'** with other objects in the interaction zone upon expulsion from outlet **314C** may cause further comminution of solid materials **12** within permeated slurry **36'** without any further energy being used.

(81) Apparatus **300** optionally includes an expansion vessel **333** including a chamber **333A** into



which permeated slurry **36'** is expelled from first chamber **314A**. Permeated slurry **36'** may be expelled into expansion vessel **333** by gravity and/or pressure differential between first chamber **314A** and expansion chamber **333A**.

(82) Working fluid (e.g. CO.sub.2) may be collected from expansion vessel **333** by a working fluid recovery system **335** which may, for example, comprise a suitable pump and/or compressor. Recovered working fluid may be conditioned in a working fluid conditioner **337** before being returned to working fluid source **316A**. The conditioning may, for example comprise removing particulates, removing contaminants, pressurizing and/or temperature adjustment. The functions of working fluid recovery system **335** and working fluid conditioner **337** may be combined.

(83) Treated slurry may be delivered from expansion chamber **333A** to a liquid/solid separation system **339** which separates the treated slurry into fractured solids **12'** and liquid **34** (e.g. water). Separated liquid **34** may be treated in a liquid treatment system **341**. Treatment system **341** may, for example perform one or more of removing particulates (e.g. by filtering, settling and/or centrifugation), degassing, removing contaminants (e.g. carbonic acid) and adjusting temperature. Treated liquid **34** may be released and/or recycled to make more slurry **36** for input to apparatus **300**.

(84) In some embodiments, if fractured solids **12'** are not sufficiently comminuted by method **200**, fractured solid materials **12'** can be returned to slurry **36** to undergo further processing according to method **200**. For example, in some embodiments solid materials **12'** (or permeated slurry **36'**) are filtered or screened after step **212** to remove undesirably large pieces of solid materials **12'** and such undesirably large pieces of solid materials **12'** may undergo further processing according to method **200**.

(85) As suggested herein, method **200** and apparatus **300** may be operated continuously (e.g. steps of method **200** can occur concurrently and in parallel). For example, new slurry **36** can be continuously prepared at step **202** and delivered through inlet **314B** at step **204**, while working fluid **16** is allowed to permeate into slurry **36** previously delivered into first chamber **314A** at step **208** and while permeated slurry **36'** is expelled from first pressure vessel **314** at step **210**. In this way, method **200** and apparatus **300** may allow for a high throughput of solid materials **12** as compared to methods which require batch processing.

(86) In some embodiments, rather than prepare slurry at step **202**, slurry is prepared within first chamber **314A** after step **208** and prior to step **210**. In such an embodiment, solid materials **12** may be delivered into first pressure vessel **314** in a similar manner as in step **102** of method **100** (and first pressure vessel **314** may be suitably modified to accommodate such delivery of solid materials **12** into first pressure vessel **314**).

(87) FIG. 5 shows schematically an example apparatus **500** according to an embodiment of the present invention. Apparatus **500** includes plural material fracturing units **514-1** to **514-N** (generally and collectively units **514**).

(88) Each unit **514** includes one or more pressure vessels which may be as described elsewhere herein. For example, each unit **514** may include: a first pressure vessel **14** like that shown in FIG. 2; a first pressure vessel **14** accompanied by a second pressure vessel **28**; as shown in FIG. 2; a first pressure vessel **314** as shown in FIG. 4, a first pressure vessel **314** accompanied by an expansion vessel **333** as shown in FIG. 4; parallel and/or serially connected groups of two or more of these etc.

(89) In some embodiments, each unit **514** includes a sequence of two or more pressure vessels in which material may be fractured by methods as described herein. In such cases the material may be passed through the two or more pressure vessels in sequence and may receive processing in each of the pressure vessels before the material exits the unit **514**.

(90) In some embodiments, each unit **514** includes one or more expansion vessels into which fractured material may be delivered from a pressure vessel. In some such embodiments where the unit **514** includes plural pressure vessels each of the plural pressure vessels may receive processed

(e.g. fractured) material from a corresponding one of the plural pressure vessels.

(91) Units **514** may be operated in sequence according to methods as described herein to maintain a more or less constant output of fractured solid materials to a downstream processing plant **504**.

Units **514** are each connected to receive a working fluid from a high pressure supply **505** of a working fluid (e.g. supercritical CO.sub.2). High pressure supply **505** may be maintained at a suitable pressure by pumps and/or compressors **506** which draw working fluid from a low pressure supply **508**, pressurize the working fluid and deliver the pressurized working fluid to high pressure supply **505**. Working fluid may be conditioned (e.g. to remove particulates and/or contaminants, to alter its temperature, to alter its pressure and/or to alter its physical phase) before being delivered into high pressure supply **505**.

(92) Apparatus **500** comprises an input material distribution system **501** which is operable to deliver solid materials **12** (e.g. rocks that have been graded and/or subjected to preliminary crushing to a size range suitable for processing in system **500**) to any of units **514**.

(93) Apparatus **500** includes an output material distribution system **502** which is operable to receive fractured solid materials **12'** that have been processed in pressure vessels **514** and to deliver fractured solid materials **12'** to material store **503** from where fractured materials **12'** may be processed in processing plant **504**.

(94) Apparatus **500** may, for example, be operated according to a method comprising: a) placing a solid material to be fractured such as rocks (optionally together with a filler material) into a pressure vessel of a unit **514**; b) delivering high pressure working fluid (e.g. in the form of one or more bursts of high pressure working fluid such as CO.sub.2) from high pressure supply **505** to the pressure vessel in the unit **514**. This may be done by controlling valves in valve bank **505A** to direct the high pressure working fluid to the desired pressure vessel of the unit **514**; c) subsequently recovering the working fluid from the unit **514**; This may be done by controlling valves in valve bank **508A** to collect working fluid from the desired pressure vessel of the unit **514**; d) conditioning the recovered working fluid to e.g. remove entrained solids and/or other contaminants and delivering the conditioned working fluid into low pressure supply **508**; e) removing the fractured solid material from unit **514**. any particular one of units **514** may be used to process solid materials at times that are staggered such that each of a plurality of units **514** may be at different stages in the above method at any given time.

(95) In some embodiments at least some of the working fluid for step b) is delivered to a pressure vessel in a unit **514** from another pressure vessel. The other pressure vessel may be in the same unit **514** or a different unit **514**. For example, step c) of the above method may include rapid decompression of a pressure vessel in a unit **514** through a fast valve, and directing a first portion of the high speed and high pressure working fluid exiting the pressure vessel into a pressure vessel in another one of units **514** that is taking in working fluid at step b) of the above method. In some embodiments step c) of the above method has plural stages. In one stage pressurized working fluid may be delivered into another pressure vessel of system **500** as described above. In another stage remaining working fluid from the pressure vessel may be directed to low pressure supply **508**.

(96) Apparatus according to the current invention (including apparatus **500**) may include a process control system (not shown) that coordinates the operation of the apparatus, for example to perform methods as described above. The process control system may, for example, operate valves, monitor conditions, operate material handling systems etc.

(97) In apparatus **500** and other embodiments as described herein) it may be beneficial to cause rapid pressure changes within pressure vessels (e.g. pressure vessels of units **514**). This may be facilitated by one or more of: locating high pressure supply **505** close to the pressure vessels of units **514**; providing a high pressure supply **505** that provides one or more accumulators located close to each unit **514**; locating low pressure reservoir(s) **508** close to the pressure vessels of units **514**; designing piping that carries working fluid from high pressure supply **505** to pressure vessels in units **514** and/or piping that carries working fluid from pressure vessels in units **514** to low

pressure reservoir **508** to offer low flow resistance (e.g. by using large diameter piping and/or piping that includes plural pipes to deliver the working fluid into or take working fluid from individual pressure vessels); using low resistance valves to control flow of the working fluid; locating valves for controlling the flow of the working fluid close to ports by way of which the working fluid is delivered into or taken from pressure vessels of units **514**; etc.

(98) In apparatus as described herein that includes two or more pressure vessels for processing materials **12**, it is an option to operate different ones of the pressure vessels to process materials out of phase with one another. This permits savings in operating expenses and capital costs, for example, by: allowing pressurized working fluid that is vented from one pressure vessel to depressurize the pressure vessel to be used directly to pressurize another one of the pressure vessels; and/or allowing supporting equipment such as expansion/collection chambers, high pressure supply of working fluid etc. to be shared among two or more of the pressure vessels.

(99) In apparatus as described herein, pressure vessels and vessels for collecting vented working fluid may be connected in a geometrical arrangement in which each pressure vessel is in close proximity to a source of high pressure working fluid and a vessel into which the working fluid may be discharged. Energy may be conserved by direct venting of pressurized working fluid from one of the pressure vessels into another one of the pressure vessels. The direct venting may be by way of connections that provide low impedance to the flow of pressurized working fluid.

(100) In any methods as described herein it may be desirable to process rocks or other irregularly shaped pieces of solid material. In such cases it can be beneficial to reduce the amount of void space in a pressure vessel. This may be done, for example, by introducing filler material into the void spaces. The filler materials may for example, comprise metal balls or other pieces of metal small enough to fit into the void spaces. The filler material may be added to solid materials being introduced into the pressure vessel (e.g. in step a) of the above method) and may be removed from the fractured solid material being removed from the pressure vessel (e.g. in step e) of the above method). The filler material may be continuously reused.

#### Interpretation of Terms

(101) Various embodiments of apparatus and/or their components described herein may use similar reference numerals (e.g. with a preceding digit, a trailing symbol, a trailing letter and/or a trailing number) to those used to describe apparatus **10** and/or its components. Unless the context or description dictates otherwise, such apparatuses and/or their components may exhibit features and/or characteristics which may be similar to the features and characteristics of apparatus **10** and/or its components (or vice versa). For example, unless the context or description dictates otherwise, first pressure vessel **314** may have features and/or characteristics similar to those discussed herein for first pressure vessel **14** (or vice versa). Further, unless the context or description dictates otherwise, it should also be understood that when referring to features and/or characteristics of apparatus **10** and/or its components, the corresponding description should be understood to apply to any of the particular embodiments of apparatuses described herein and/or their components.

(102) Unless the context clearly requires otherwise, throughout the description and the claims: “comprise”, “comprising”, and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to”; “connected”, “coupled”, or any variant thereof, means any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof; elements which are integrally formed may be considered to be connected or coupled; “herein”, “above”, “below”, and words of similar import, when used to describe this specification, shall refer to this specification as a whole, and not to any particular portions of this specification; “or”, in reference to a list of two or more items, covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list; the singular forms “a”, “an”, and “the” also include the

meaning of any appropriate plural forms.

(103) Words that indicate directions such as “vertical”, “transverse”, “horizontal”, “upward”, “downward”, “forward”, “backward”, “inward”, “outward”, “left”, “right”, “front”, “back”, “top”, “bottom”, “below”, “above”, “under”, and the like, used in this description and any accompanying claims (where present), depend on the specific orientation of the apparatus described and illustrated. The subject matter described herein may assume various alternative orientations. Accordingly, these directional terms are not strictly defined and should not be interpreted narrowly.

(104) Unless the context clearly requires otherwise, throughout the description and the claims, reference to a “pump” or “pumping” are to be construed to include compressors and compressing and any other apparatus or method of increasing the pressure of a fluid (e.g. working fluid **16**).

(105) Embodiments of the present invention include various operations, which are described herein. These operations may be performed by hardware components, software, firmware, or a combination thereof.

(106) Although the operations of the method(s) herein are shown and described in a particular order, the order of the operations of each method may be altered so that certain operations may be performed in an inverse order or so that certain operation may be performed, at least in part, concurrently with other operations. In another embodiment, instructions or sub-operations of distinct operations may be in an intermittent and/or alternating manner.

(107) Where a component (e.g. a software module, processor, assembly, device, circuit, etc.) is referred to above, unless otherwise indicated, reference to that component (including a reference to a “means”) should be interpreted as including as equivalents of that component any component which performs the function of the described component (i.e. that is functionally equivalent), including components which are not structurally equivalent to the disclosed structure which performs the function in the illustrated exemplary embodiments of the invention.

(108) Specific examples of systems, methods and apparatus have been described herein for purposes of illustration. These are only examples. The technology provided herein can be applied to systems other than the example systems described herein. Many alterations, modifications, additions, omissions, and permutations are possible within the practice of this invention. This invention includes variations on described embodiments that would be apparent to the skilled addressee, including variations obtained by: replacing features, elements and/or acts with equivalent features, elements and/or acts; mixing and matching of features, elements and/or acts from different embodiments; combining features, elements and/or acts from embodiments as described herein with features, elements and/or acts of other technology; and/or omitting combining features, elements and/or acts from described embodiments.

(109) Various features are described herein as being present in “some embodiments”. Such features are not mandatory and may not be present in all embodiments. Embodiments of the invention may include zero, any one or any combination of two or more of such features. This is limited only to the extent that certain ones of such features are incompatible with other ones of such features in the sense that it would be impossible for a person of ordinary skill in the art to construct a practical embodiment that combines such incompatible features. Consequently, the description that “some embodiments” possess feature A and “some embodiments” possess feature B should be interpreted as an express indication that the inventors also contemplate embodiments which combine features A and B (unless the description states otherwise or features A and B are fundamentally incompatible). This is the case even if features A and B are described in different sentences, different sections or with reference to different figures of the drawings.

(110) While a number of exemplary aspects and embodiments have been discussed above, those of skill in the art will recognize certain modifications, permutations, additions and sub-combinations thereof. It is therefore intended that the following appended claims and claims hereafter introduced are interpreted to include all such modifications, permutations, additions and sub-combinations as are consistent with the broadest interpretation of the specification as a whole.

## Claims

1. A method for comminution of solid materials, the method comprising: introducing solid materials into a first pressure vessel; providing a working fluid in the first pressure vessel at an operating pressure and an operating temperature; allowing the working fluid to permeate into the solid materials; and rapidly expanding the working fluid to thereby create fractured solid materials from the solid materials; wherein the step of introducing the solid materials into the first pressure vessel comprises mixing the solid materials with a liquid to form a slurry and introducing the slurry into the first pressure vessel; and wherein the step of allowing the working fluid to permeate into the solid materials comprises separating the solid materials from the slurry and allowing the working fluid to permeate into the separated solid materials.
2. The method according to claim 1 wherein the step of rapidly expanding the working fluid comprises reducing the pressure of the working fluid from the operating pressure to a second pressure lower than the operating pressure.
3. The method according to claim 2 wherein the step of reducing the pressure of the working fluid from the operating pressure to the second pressure comprises reducing the operating pressure over a time period that is less than 30 ms.
4. The method according to claim 1 wherein the solid material comprise porous ore-bearing rock.
5. The method according to claim 1 wherein the working fluid comprises CO.sub.2.
6. The method according to claim 1 comprising re-combining the separated solid materials and the liquid to reform the slurry before rapidly expanding the working fluid to thereby create the fractured solid materials from the solid materials.
7. The method according to claim 1 wherein the step of rapidly expanding the working fluid to thereby create the fractured solid materials from the solid materials comprises expelling the slurry from the first pressure vessel into an environment having a pressure that is lower than a pressure within the first pressure vessel.
8. The method according to claim 7 wherein the step of expelling the slurry from the first pressure vessel comprises expelling the slurry into an expansion chamber and collecting the working fluid from the expansion chamber.
9. The method according to claim 7 comprising directing the expelled slurry at a solid surface to cause further pulverization of the fractured solid materials.
10. The method according to claim 7 comprising directing the expelled slurry to impinge against a medium containing additional fractured solid materials to cause further pulverization of the fractured solid materials.
11. The method according to claim 10 wherein the medium comprises a pool of the slurry.
12. The method according to claim 2 wherein the step of reducing the pressure of the working fluid to the second pressure comprises opening a valve to release at least some of the working fluid through the valve into a second pressure vessel for recapture.
13. The method according to claim 2 comprising after the step of rapidly expanding the working fluid and before removing the fractured solid materials from the first pressure vessel: increasing the temperature of the working fluid in the first pressure vessel to the operating temperature; increasing the pressure of the working fluid in the first pressure vessel to the operating pressure and rapidly expanding the working fluid to further fracture the fractured solid materials.
14. The method according to claim 13 wherein the step of increasing the pressure of the working fluid to the operating pressure comprises injecting working fluid recaptured in a second pressure vessel into the first pressure vessel, wherein the recaptured working fluid is recaptured during at least one of the steps of rapidly expanding the working fluid.
15. The method according to claim 14 comprising conditioning the recaptured working fluid by one of more of: compressing the working fluid, pressurizing the working fluid, liquefying the working

fluid, removing solid particles from the working fluid, filtering the working fluid and removing containments from the working fluid prior to injecting the recaptured working fluid into the first pressure vessel.

16. The method according to claim 1 wherein the step of introducing the solid materials into the first pressure vessel comprises introducing the solid materials into the first pressure vessel through a first airlock, a first lock hopper or a first piston feed and wherein removing the fractured solid materials from the first pressure vessel comprises removing the fractured solid materials from the first pressure vessel through the first airlock, a second airlock, the first lock hopper, a second lock hopper, the first piston feed, a second piston feed, a rotary valve, a plug-forming feeder or a dynamic feeder.

17. An apparatus for comminution of solid materials, the apparatus comprising: a first pressure vessel, the first pressure vessel comprising a first chamber; a source of pressurized working fluid connected to deliver the pressurized working fluid into the first chamber through a second inlet; a second pressure vessel connected to the first chamber by a first valve, wherein when the first valve is open, at least some of the working fluid within the first chamber is allowed to flow out of a first outlet and into a second chamber of the second pressure vessel; a density sensor for measuring a density of the working fluid in the first pressure vessel; and a density controller connected to control one or both of the temperature and the pressure of the working fluid in the first pressure vessel in response to the density measured by the density sensor so as to maintain the density of the working fluid within the first pressure vessel within a desired density range.

18. A method for comminution of solid materials, the method comprising: introducing solid materials into a first pressure vessel; providing a working fluid in the first pressure vessel at an operating pressure and an operating temperature; allowing the working fluid to permeate into the solid materials; and rapidly expanding the working fluid to thereby create fractured solid materials from the solid materials; measuring a density of the working fluid in the first pressure vessel; and controlling one or both of the temperature and the pressure of the working fluid in the first pressure vessel in response to the measured density so as to maintain the density of the working fluid within the first pressure vessel within a desired density range.

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